Trafag AG · Microswitch EOL tester

ScadaViewer

Technical Manual

Web application for viewing, evaluating and exporting EOL tester data — architecture, interfaces, configuration, installation and operation

Customer	**Trafag AG**
Project	**50-1182875**
Machine	**Microswitch EOL tester**
Software	**ScadaViewer 2.7.0**
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1 About this document

This technical manual describes ScadaViewer version 2.7.0: architecture, data sources, PLC interface, users and access, REST API, configuration, installation, operation and troubleshooting. Operation of the application is described in the ScadaViewer Instructions for Use.

Target group: EOL tester service and maintenance, IT administrator, PLC programmer, developer.

Conventions:

> **Caution:** Notice of a situation that may lead to data loss, incorrect evaluation or equipment damage.

> **Note:** Additional information for understanding the text.

> **Tip:** Recommended practice or practical advice.

Screen texts (buttons, messages) are shown in **bold** or in quotation marks, file, column and value names in `monospace`.

Related documents:

- **ScadaViewer — Instructions for Use**: operating the application.
- **DatabaseGateway — Technical Manual**: file writing and NAS synchronisation (same data folder and same NAS).
- **Analyzing — Technical Manual**: calculation of the parameters shown by ScadaViewer.

Table 1 – Revision history

Issue	Date	Change
1.0	2026-10-28	First issue.
1.1	2026-10-29	Update for software version 2.7.0 (final release).

2 Operating principle

ScadaViewer runs on the tester PC as the Windows service `ScadaViewer` (program `scada_viewer.exe`). One process serves the web server (port 8080), REST API, WebSocket and the web user interface (React). It reads data from the CSV files written by DatabaseGateway and the tester state from the PLC via ADS. On the tester PC the application is opened in a kiosk at `http://localhost:8080`, from the company network in a browser.

2.1 Components

Table 2 – ScadaViewer components

Component	Task
Web server (FastAPI, uvicorn)	REST API `/api/*`, WebSocket `/ws/plc`, static files of the web interface; GZip, security headers, request limit.
ADS monitor	Notifications of 24 PLC symbols → WebSocket; heartbeat and readiness to the PLC every 0.5 s; reconnection after an outage (pause 1 → 30 s).
File watcher	Every 2 s compares the content of the local folders; on a change it sends “files_changed” to the browsers (automatic Database refresh).
File overview and cache	Order overview from file names, piece counts calculated in the background, metadata cache in SQLite (`03_output\cache\file_meta.sqlite`).
NAS access	Own pool of 6 threads; full → “NAS busy” response (503); NAS availability and usage are checked by one thread with a 60 s cache.
Web interface	React application (Database, Detail, Settings, Info pages); communicates only via the REST API and WebSocket.

2.2 Data sources

Table 3 – Data sources

Source	Path	Access
Local data (tester PC)	`{local_path}\production|testing\wip|done_local|done_remote`	read; deleting and finishing an order only on user request at the tester
NAS	`{remote_path}\production`, `{remote_path}\testing`	read only — never writes or deletes on the NAS; ignores `*.part` files
PLC	`GV_IO_ADS_API.ScadaViewerApp`	ADS notifications; writes only heartbeat and readiness
USB flash drive	`X:\Trafag_ScadaViewer`	export writing (administrator at the tester only)

The file state is derived from the folder: `wip` = In progress, `done_local` = Local (waiting for the NAS), `done_remote` = Synced. The file format is described in the DatabaseGateway Technical Description.

2.3 Faults and outages

Table 4 – Behaviour during faults

Situation	Behaviour
PLC / ADS outage	ADS state red in the top bar, symbol values are discarded (a new client does not receive old data), the PLC sign-in ends; repeated reconnection.
NAS unavailable	The Remote tab reports unavailability; local data, PLC and the rest of the application keep running.
NAS hangs / is slow	NAS operations fail after 30 s; at most 6 concurrent operations, further ones get 503 immediately.
`local_path` folder missing	The application runs in the "degraded" state (`/api/health`).
Order above 200 MB	Order detail refused (413) — data available only by downloading the original.

3 PLC interface

Communication via the ADS protocol (TCP 48898) with the PLC runtime (AMS Net ID and port from `[ads]`), global variable list `GV_IO_ADS_API.ScadaViewerApp`. The values of the `Out` symbols arrive as ADS notifications on change.

Table 5 – Interface symbols

Symbol (under the GVL)	Type	Direction	Meaning
`Out.Status.Mode`	UINT	PLC → SV	Machine mode (E_APP_ModeManager_Mode)
`Out.Status.Order.Valid / Name / Count_Expected / Count_Actual`	BOOL / STRING / UINT	PLC → SV	Running order and piece counts
`Out.Status.UserLoggedIn`	BOOL	PLC → SV	Sign-in on the PLC (section 4.3)
`Out.Status.Boxes.Presence / Full / Count [1..6]`	BOOL / BOOL / UINT	PLC → SV	Sorting box state
`In.Status.Heartbeat`	BOOL	SV → PLC	Toggles every 0.5 s
`In.Status.Ready`	BOOL	SV → PLC	ScadaViewer ready

An order reported as running by the PLC (`Order.Valid` and `Order.Name`) cannot be finished manually in ScadaViewer (409).

4 Users and access

4.1 Roles

Table 6 – User roles

Role	At the tester	Remotely
operator	viewing, printing	viewing, CSV / XLSX export to the browser, printing
technician	+ deleting synchronised files	as operator
admin	+ export to USB, finishing / deleting an order in progress, deleting Local files, PC time, paths and storage limit, user management	as operator + own password
manufacturer	everything	as operator

Clean-up of synchronised files is allowed for every signed-in user at the tester. Accounts are in `users.toml` next to `Config.toml`; until the file exists, the single account from `Config.toml [auth]` applies.

4.2 At the tester and remotely

The mode is decided by the address the connection comes from: addresses in `[server] local_clients` (default `127.0.0.1`, `::1`) = **at the tester**, others = **remote access** (viewing only). Write actions additionally require `localhost` or an IP address in the Host header. Proxy headers (`X-Forwarded-For`) are ignored.

> **Caution:** Do not add the tester network to `local_clients` — a laptop with a static address would get full rights and automatic sign-in from the PLC. The kiosk must open `http://localhost:8080`, not the IP address of the PC.

4.3 Sign-in and sessions

- **PLC sign-in** — at the tester only: if a user with the highest authority is signed in on the PLC (`UserLoggedIn`), ScadaViewer signs them in with the admin role. The session is valid as long as the flag is set; signing out on the PLC or an ADS outage ends it.
- **Password** — the session is valid for 8 h; at the tester it ends after 15 min of inactivity. 5 wrong passwords block sign-in from that address for 10 min.
- **Default password** `admin` / `admin`: while any account has it, the application shows a warning, `/api/health` reports `default_password: true` and remote sign-in is refused.
- Limit of 120 requests per minute per address (the kiosk at the tester is exempt).

5 REST API and WebSocket

Protected endpoints require `Authorization: Bearer <token>` (from `POST /api/auth/login`); an invalid or expired token = 401 with the `WWW-Authenticate: Bearer` header. Write endpoints return 403 remotely. The full parameter description is in the Swagger UI at `http://localhost:8080/docs`.

Table 7 – API overview

Endpoint	Purpose
`GET /api/health`	Application status (no sign-in) — section 8.1
`POST /api/auth/login | logout | change-password | plc-login`	Sign-in, sign-out, password change, PLC sign-in (tester only)
`GET /api/files`	List of orders / tests (`location`, `type`, paging, sorting, `q` search, `from` / `to`) + orders in progress
`GET /api/files/{id}/download | /xlsx`	Original CSV / XLSX built by the server
`DELETE /api/files/{id}`, `POST /api/files/batch-delete`, `POST /api/files/{id}/finish`	Deleting and manually finishing an order in progress (at the tester, by role)
`GET /api/data`	Order pieces / test data with filters (`status`, `category`, nok, `order`, paging; limit 20,000 rows)
`GET /api/signal`, `GET /api/timeline`	Test signal data (decimated, range), order timeline
`GET /api/status`, `GET /api/storage`, `POST /api/storage/cleanup`	NAS availability, storage usage, clean-up of synchronised files
`/api/usb*`	USB flash drive: list, export, bulk save, safe removal (admin at the tester)
`POST /api/bulk/plan | ticket`, `GET /api/bulk/zip`	Bulk download as ZIP (max. 500 files, one-time link)
`/api/users*`, `/api/config*`, `/api/system/time`	User management, configuration (paths, storage limit), PC time and NTP
`WS /ws/plc`	Live PLC symbol values, ADS state, “files_changed” (no sign-in)

6 Configuration (Config.toml)

`Config.toml` is in the application folder (template `Config.toml.example`). The configuration is loaded at start-up; data paths and the storage limit can also be changed in Settings (effective immediately), other changes require a service restart.

Table 8 – Config.toml keys

Key	Example	Meaning
`[server] host`	`"0.0.0.0"`	all network interfaces (tester network and company network)
`[server] port`	`8080`	application port
`[server] cors_origins`	`["*"]`	allowed addresses for the WebSocket; `["*"]` recommended with DHCP
`[server] local_clients`	`["127.0.0.1", "::1"]`	"at the tester" addresses (section 4.2)
`[ads] net_id` / `port`	`"5.12.34.56.1.1"` / `851`	AMS Net ID and port of the PLC runtime
`[data] local_path`	`"C:/Data/DtbLoc"`	DatabaseGateway data folder (= its `local_base_dir`)
`[data] remote_path`	`"\\\\10.0.0.20\\scada_archive"`	UNC path to the NAS (every \ as \\); empty = NAS not used
`[data] csv_separator` / `csv_encoding`	`";"` / `"utf-8-sig"`	same as DatabaseGateway
`[data] local_max_gb`	`5.0`	local storage limit; ≥ 80 % warning, ≥ 95 % critical
`[auth] username` / `password_hash`	`"admin"`	account valid until `users.toml` exists

> **Note:** NAS: `remote_path` must match DatabaseGateway (`\\{host}\{share}`; its `production_path` and `testing_path` = `{remote_path}\Production` and `\Testing`). Write the NAS address the same way in both applications (IP or DNS name) and always as a UNC path, never a drive letter.

7 Installation and update

7.1 Package contents

Table 9 – Package contents

File / folder	Content
`scada_viewer.exe`	Launcher — does not work alone, always with `_internal`
`_internal\`	Python, libraries (FastAPI, pyads, numpy) and the web interface
`Config.toml.example`	Configuration template (the package contains no `Config.toml`)
`nssm_install.bat`	Windows service installation
`kiosk_start.bat`, `open_browser.bat`, `start.bat`	Kiosk (2 screens: ScadaViewer + HMI), opening the browser, manual start
`03_output\logs\`	Log folder

7.2 First installation

1. Extract the package (e.g. `C:\apps\…-ScadaViewer\scada_viewer`) and create `Config.toml` from `Config.toml.example` (chapter 6).

2. Place `nssm.exe` (64-bit, nssm.cc) in the application folder or on the PATH and run `nssm_install.bat` as administrator — service `ScadaViewer`, automatic start, logs `03_output\logs\nssm_*.log` rotated every 10 MB.
3. For NAS access run the service under the same Windows service account as DatabaseGateway and store the NAS credentials for this account with `cmdkey` (the same NAS account as DatabaseGateway — otherwise error 1219).
4. Firewall: allow port 8080 only from the company network (or specific PCs); the kiosk uses localhost.
5. Open `http://localhost:8080`, sign in as `admin` / `admin`, create personal accounts in Settings → Users and change the password of the admin account.

```
runas /user:.\svc_tester "cmdkey /add:10.0.0.20 /user:<účet NAS> /pass:<heslo>"
netsh advfirewall firewall add rule name="ScadaViewer" dir=in action=allow protocol=TCP
localport=8080 remoteip=10.0.0.0/16
```

7.3 Update on the tester PC

The tester applications (Analyzing, DatabaseGateway, ScadaViewer) are updated together with the deployment package `deploy_v<version>` (`deploy.bat` as administrator, roll back with `rollback.bat`). The script stops the services, replaces `exe` and `_internal` (backup `<folder>.backup_<time>`), keeps `Config.toml` and `users.toml`, starts and verifies the services. Browsers load the new web interface automatically.

7.4 Stopping and uninstallation

```
nssm stop ScadaViewer
nssm remove ScadaViewer confirm
```

8 Operation and diagnostics

8.1 How to check the state

`GET http://localhost:8080/api/health` (always HTTP 200, no sign-in):

Table 10 – /api/health response

Field	Meaning
`status`	ok, or degraded if the `local_path` folder does not exist
`checks.local_storage`	data folder exists
`checks.ads`	connection to the PLC
`checks.default_password`	some account has the default password
`checks.users_file_error`	`users.toml` cannot be read (user changes blocked)

8.2 Logs

- `03_output\logs\nssm_stdout.log` — application progress (JSON: `ts`, `level`, `mod`, `msg`), rotated every 10 MB; rotated logs older than 30 days are deleted at start-up.
- `03_output\logs\nssm_stderr.log` — start-up errors.

- Message prefixes: `[APP]` application, `[API]` requests, `[ADS]` PLC, `[CSV]` files, `[SVC]` services, `[WS]` WebSocket, `[AUTH]` sign-in and refused remote writes.
- Manual run with detailed output: `scada_viewer.exe --config Config.toml --debug` (only with the service stopped — it connects to the PLC).

8.3 Cache and memory

- `03_output\cache\file_meta.sqlite` (+ `-wal`, `-shm`) — file metadata and piece counts; can be deleted at any time with the service stopped (recalculated).
- The NAS listing is kept in memory for 15 s; open orders at most 3 files / 256 MB; piece counts of NAS files are calculated only for the displayed page.
- Service memory approximately 0.3–0.6 GB (≈ 1 kB per file in the overview).

9 Troubleshooting

Table 11 – Troubleshooting

Symptom	Cause	Solution
Service does not start	missing `Config.toml`, syntax error (\ instead of \\), invalid `local_clients` entry, port 8080 in use (WinError 10048)	`nssm_stderr.log`; manual run with `--config`
ADS state red	wrong `net_id`, TwinCAT not in RUN, firewall (TCP 48898)	TwinCAT → Routes; `sc query TcSystemService`
NAS unavailable	service under LocalSystem without NAS credentials, missing / invalid `cmdkey`, drive letter in `remote_path`, TCP 445 blocked	`runas /user:.\svc_tester "cmd /k dir \\10.0.0.20\scada_archive"`
Remote tab does not show files uploaded by Gateway	`remote_path` does not match DatabaseGateway `production_path` / `testing_path`	Align the paths (chapter 6)
"…" instead of piece count	the count is calculated in the background (from the NAS only the displayed page)	Wait; if it takes long, check the NAS and the `[CSV]` log
Page does not open from the office	firewall, VLAN, proxy exception, IP change (DHCP)	Rule for 8080, DNS name or DHCP reservation
Remotely the PLC state and automatic refresh do not work	`cors_origins` does not contain the address used	`cors_origins = ["*"]`
PLC sign-in and deleting missing at the tester	the kiosk opens the IP address instead of `localhost`	Fix the address in `kiosk_start.bat`
Administrator password forgotten	—	Stop the service, replace `password_hash` in `users.toml` with a new hash (PBKDF2-SHA256, 260,000 iterations, `salt:hash`), start the service

Appendix A Terms and abbreviations

Table 12 – Terms and abbreviations

Term	Meaning
EOL tester	Microswitch EOL (end-of-line) tester — test station at the end of production; “tester” for short in the text.
Tester PC	Industrial PC of the EOL tester running Analyzing, DatabaseGateway and ScadaViewer. “PC at the machine” in the application texts.
ADS / AMS Net ID	Beckhoff TwinCAT communication protocol / device address in the ADS network.
NAS / UNC	Synology network storage / Windows network path `\\server\folder`.
NSSM	Tool running a program as a Windows service.
Kiosk	Full-screen browser on the tester PC without browser controls.
REST API / WebSocket	HTTP interface for data / persistent connection for live values.